

DL CHROME EXTENSION

DataLemur Sync

User guide & troubleshooting — save your accepted SQL solutions to GitHub automatically.

Version 1.0.0 · Manifest V3 · No backend server · Your token never leaves this computer

1. What this extension does

When you solve a question on DataLemur, this extension reads the problem name, its difficulty and the SQL in the editor, then commits it to a GitHub repository you own — one file per problem, sorted into `Easy/`, `Medium/` and `Hard/` folders, with the repository's `README.md` kept up to date.



Solving the same problem again updates the existing file — it never creates a duplicate.

2. One-time setup

1 Build the extension

Open a terminal in the project folder `C:\SQL_Sync` and run `npm install` followed by `npm run build`. This creates the `dist` folder — that folder is the extension.

2 Load it into Chrome

Go to `chrome://extensions`, turn on **Developer mode** (top-right switch), click **Load unpacked**, and select `C:\SQL_Sync\dist`. Then open the puzzle-piece menu in the toolbar and pin **DataLemur Sync** so its icon is always visible.

3 Create the destination repository

On github.com create a new repository, for example `datalemur-solutions`. Tick **Add a README file** when creating it. A completely empty repository has no branch yet and the extension cannot write to it.

4 Create a Personal Access Token

Go to github.com/settings/tokens → *Generate new token (classic)* → tick the `repo` scope. Copy the token immediately — GitHub shows it only once. A fine-grained token also works if you give it *Contents: Read and write* on that repository.

5 Fill in the settings page

The settings page opens by itself after installing. To reopen it later, right-click the extension icon and choose **Options**. Enter your token, GitHub username, repository name and branch (usually `main`), then press **Save**.

6 Press "Test connection"

You should see ✓ **Connected — the repository is writable**. If you see anything else, fix it now using the troubleshooting table before going to DataLemur.

IMPORTANT

Save and **Test connection** are two different buttons. Testing does not save. If you only test and then close the page, your settings are gone.

3. Everyday use

1 Open a question on DataLemur

Log in and open any SQL question. The address must look like `datalemur.com/questions/some-question-name`. The extension stays inactive on the question list, the blog and every other site — that is deliberate.

2 Click the extension icon

A small panel opens. Even before you submit anything, it already reads the page:

Status (top-right)	Not accepted yet — sync anyway
Problem	Histogram of Tweets
Difficulty	Easy
Repository	yourname/datalemur-solutions (main)
Path	Easy/Histogram of Tweets.sql

THIS IS NORMAL

The orange "Not accepted yet" message is not an error. It only means you have not yet submitted a correct answer on this page. Everything else is already working.

3 Write your query and submit it on DataLemur

Use the website's own Run / Submit button. The extension is not involved in this step.

4 Wait for the green tick

When DataLemur reports your answer is correct, a small green ✓ badge appears on the extension icon, and the panel's status changes to ✓ **Ready to Sync**.

5 Optional: press Preview

This shows the exact file that will be committed — three comment lines (problem name, difficulty, link back to the question) followed by your query. Press **Preview** again to hide it.

6 Press Sync

A notification confirms ✓ **Solution Synced**. Open your repository on github.com: the `.sql` file is in its difficulty folder and `README.md` has been rewritten with your new totals.

What each panel button does

Button	Effect
Sync	Commits the current solution to GitHub and updates the repository README. Disabled until settings are complete and SQL has been detected.

Preview	Shows exactly what will be committed. Reads nothing from the internet and changes nothing.
Settings	Opens the options page (token, username, repository, branch, auto-sync, developer logging).

Automatic syncing

In Settings, tick **Sync automatically when a submission is accepted**. From then on you never press Sync — the moment DataLemur accepts an answer, it is committed. The same solution is never committed twice, so re-rendering the page cannot create repeat commits.

The Progress panel

The bottom of the panel counts what *this browser* has synced: total, Easy, Medium, Hard and the time of the last sync. It is a local counter, not a reading of your repository. **Reset progress** in Settings clears these numbers and never touches anything on GitHub.

4. Troubleshooting

Messages in the panel

What you see	What it means and what to do
Open a DataLemur question	You are not on a <code>/questions/<name></code> page — or the tab was already open when you installed or reloaded the extension. Reload the tab (F5) and try again.
Waiting for the editor...	The page loaded but the SQL editor has not appeared yet, or it is empty. Type something into the SQL box, then close and reopen the panel.
Settings required	Token, username, repository or branch is missing. Open Settings, complete every field and press Save .
Not accepted yet — sync anyway	Normal before you submit a correct answer. Sync still works if you press it — useful if DataLemur changes its result banner and the tick stops appearing.
Sync button greyed out	Either no SQL was detected on the page, or the settings are incomplete. Check both.
Already exists	Not an error. The file on GitHub is already identical to your current solution, so nothing was committed.

Errors reported as notifications

Message	Cause and fix
Invalid or expired GitHub token	The token is mistyped, was revoked, or has expired. Generate a new one and paste it again. Make sure there is no trailing space.
Repository not found	One of three things: the username or repository name is spelled wrong; the branch name is wrong (<code>main</code> vs <code>master</code>); or the repository has no commits yet. Open the repo on github.com and copy the exact values from the address bar.
Permission denied	The token is valid but not allowed to write. Classic tokens need the <code>repo</code> scope; fine-grained tokens need <i>Contents: Read and write</i> and must list this repository under repository access.
GitHub rate limit reached	Too many requests in a short window. Wait a few minutes and sync again.
Network error	No internet, a VPN or a firewall blocking <code>api.github.com</code> . The extension already retried three times before showing this.
Something went wrong while syncing	An unexpected response from GitHub. Open the service worker console (see below) — the exact reason is printed there.

Wrong information was detected

Symptom	Explanation
Difficulty says Medium but the question is Easy	The difficulty label could not be read, so the extension used a safe default rather than refusing to sync. The file is committed correctly, just in the wrong folder — move it on GitHub.
The problem name looks like a URL slug	The title could not be read from the page, so the extension built one from the address. The sync still works.
The green tick never appears after a correct answer	DataLemur changed the wording of its result banner. Press Sync manually — it is never blocked by this.

DESIGN NOTE

The extension is built to degrade rather than fail. Whenever it cannot read something from the page, it falls back to a sensible default and still lets you sync — you never lose a solution because a website changed its layout.

Looking at the logs

Open Settings and tick **Developer logging in the console**. Every message is prefixed with `[DataLemur Sync]`. There are two separate consoles:

- **Page console** — press F12 on the DataLemur tab. Shows what was read from the page: `Extracting DOM...`, `Parsed problem {...}`, `Submission accepted`.
- **Extension console** — go to `chrome://extensions` and click the **service worker** link on the DataLemur Sync card. Shows the GitHub side: `Verifying repository access...`, `Uploading...`, `Success`, and the detail of any failure.

The service worker goes to sleep after about thirty seconds of inactivity and its console goes quiet. That is normal Chrome behaviour — syncing wakes it up again.

After changing the code

Run `npm run build`, then press the circular reload arrow on the extension card in `chrome://extensions`, then reload the DataLemur tab. All three steps are needed: an already-open page keeps running the old copy of the extension.

5. Privacy

- Your token is stored only in this browser's local extension storage. It is never synced to your Google account and never sent to any server other than `api.github.com`.
- The token is never printed to a console, never written into a committed file, and never given to the DataLemur page.
- The extension can only read `datalemur.com` and talk to `api.github.com`. It has no other access.
- There is no analytics, no tracking and no remote code of any kind.

DataLemur Sync 1.0.0 — user guide. Project folder: C:\SQL_Sync · See [README.md](#) for developer documentation and [docs/TESTING.md](#) for the full test checklist.